How Many Minutes Until a Number Means Something? Sampling Uncertainty in Data-Driven Scouting

Introduction. Percentile charts are the common language of data scouting: a player's per-90 output ranked against his positional peers, drawn as a radar or a pizza, and shared in reports as if each slice were a fact. However, they carry no error bars, and 1,000 minutes is barely a dozen matches: every slice is an estimate built on a small sample. In this paper, we ask how much of a per-90 percentile is signal, how many matches each metric needs before it repeats itself, and how often the ranking a scout sees in October is the one he would see in May.

Methods. From 1,517 matches of the 2015/16 La Liga, Premier League, Serie A and Ligue 1 (StatsBomb open data), we built 36,052 appearances of at least 30 minutes, 2,093 player-seasons and 13 scouting metrics across seven outfield position groups. Reliability: for every metric, position and sample size n, we computed the Spearman correlation across players between two disjoint random samples of n matches, averaged over 40 splits. Minutes-to-stability is the sample size at which this correlation reaches 0.7. Percentile uncertainty: for each of the 1,313 player-seasons above 900 minutes, we resampled matches 400 times to obtain a 90% interval on the positional percentile. Ranking stability: we measured the share of each league-position top-10 at season end that was already in the top-10 after 5, 10, 15 and 20 matches, using the same players.

Results. Metrics stabilise at very different speeds. Pass completion becomes repeatable after 6 matches (about 480 minutes), pressures, progressive passes and carries after 7–8, aerial duels after 9, box touches after 10 and tackles plus interceptions after 14. Six metrics fall short of 0.7 even after 16 matches, more than 40% of a season: shots (0.67), ball recoveries (0.66), successful dribbles (0.65), npxG (0.62), key passes (0.56) and xA (0.46). The metrics that matter most in recruitment are the noisiest: after five matches, shots, npxG, key passes and xA correlate at only 0.24–0.38 with the next five (Figure 1). Percentiles inherit that noise. Between 900 and 1,500 minutes, the median 90% interval spans 31 to 57 percentile points depending on the metric, and 72% of all percentile estimates in that range have an interval of 30 points or more. Even above 2,500 minutes, xA and key passes still have intervals wider than 40 points (Figure 2). Rankings move accordingly: after five matches only 54–73% of the eventual top-10 is in place, 75% after ten and 82–93% after twenty.

Conclusion. A per-90 percentile computed on a typical scouting sample is a range, not a point, and the width of that range depends on the metric far more than on the player. We publish minutes-to-stability for 13 metrics, an interval for every percentile, and an honest version of the chart the industry already uses.

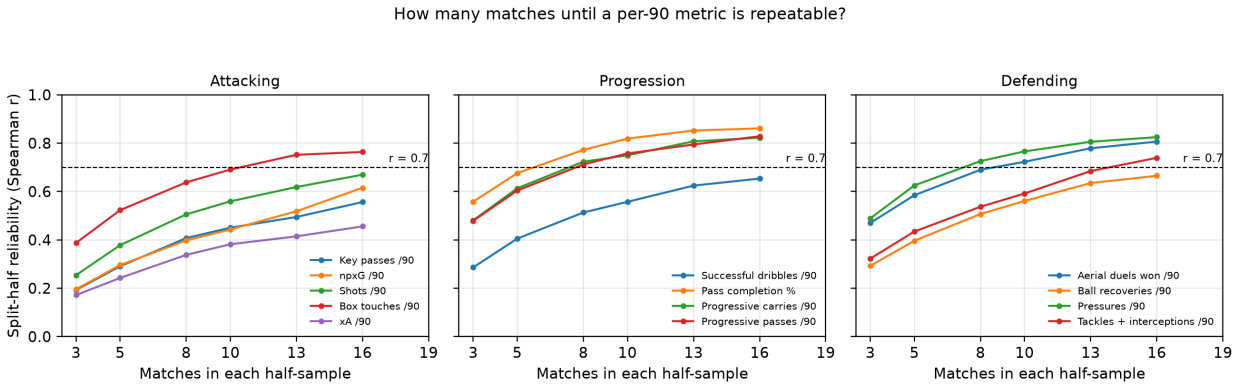


Figure 1. Split-half reliability of 13 metrics as a function of matches per sample, pooled across position groups (dashed line: $r = 0.7$).

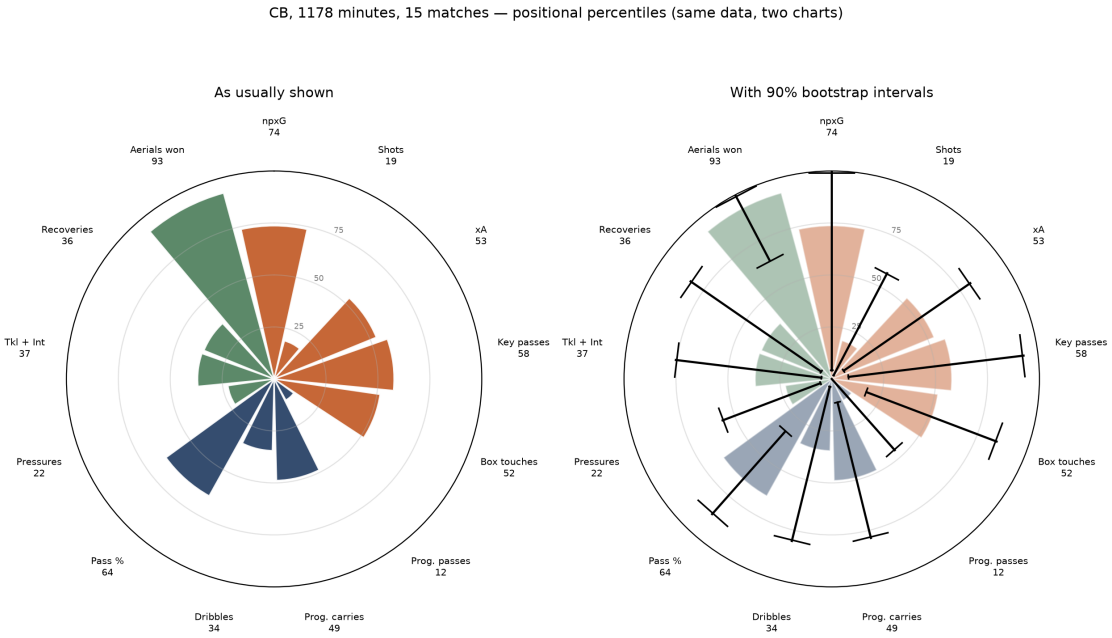


Figure 2. The same centre-back (1,178 minutes, 15 matches) as usually shown and with 90% bootstrap intervals on every positional percentile.